

# Pion EMFF Background-Response Diagnostic

Finite-Field Energy Shift versus Linearized Response Propagator

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May 20, 2026

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## 1 Purpose

This note documents the pion electromagnetic form-factor (EMFF) background-response diagnostic implemented in

`pyquda_measurement_utils/pion_EMFF_background_response_vibe_develop.py`

and

`application/EMFF_pion_background_response/perlmutter/.`

The goal is to study a background-field/Feynman-Hellmann route to pion EMFF matrix elements while *not* modifying the QUDA Dirac operator and while *not* performing inversions with a finite external field  $D + \lambda\mathcal{O}$ . Instead, the current implementation constructs the first-order response propagator using ordinary  $D^{-1}$  inversions.

This is not yet the full finite- $\lambda$  background-field method. It is the linearized method at  $\lambda = 0$ , which is mathematically equivalent to a summed current insertion.

## 2 Two Related But Distinct Methods

### 2.1 Finite- $\lambda$ background-field method

The most literal background-field method modifies the action or Dirac operator,

$$D_\lambda = D + \lambda\mathcal{O}, \quad (1)$$

then computes ordinary two-point functions in this modified theory,

$$C_\lambda(t) = \langle O_\pi(t) O_\pi^\dagger(0) \rangle_\lambda. \quad (2)$$

At large Euclidean time,

$$C_\lambda(t) = Z_\lambda e^{-E_\lambda t} + \dots. \quad (3)$$

The desired matrix element is obtained from the energy response,

$$\left. \frac{\partial E_\lambda}{\partial \lambda} \right|_{\lambda=0}. \quad (4)$$

This is the usual “extract the energy shift” picture of the background-field method.

### 2.2 Linearized response at $\lambda = 0$

The method currently implemented in this repository does not invert  $D + \lambda\mathcal{O}$ . Instead, it expands the propagator,

$$S_\lambda = (D + \lambda\mathcal{O})^{-1} = S - \lambda S\mathcal{O}S + O(\lambda^2). \quad (5)$$

Thus,

$$\left. \frac{\partial S_\lambda}{\partial \lambda} \right|_{\lambda=0} = -S\mathcal{O}S. \quad (6)$$

The code constructs the object

$$S_{\text{resp}} = S\mathcal{O}S = D^{-1}\mathcal{O}S, \quad (7)$$

using ordinary inversions. Therefore, relative to the derivative in Eq. (6), the physical finite-difference sign is

$$\left. \frac{\partial S_\lambda}{\partial \lambda} \right|_{\lambda=0} = -S_{\text{resp}}. \quad (8)$$

The positive-sign  $S_{\text{resp}}$  convention is useful because it can be compared directly against the explicit summed three-point function.

### 2.3 Why the linear response looks like a summed three-point function

Expanding the two-point function in the external source gives

$$\left. \frac{\partial C_\lambda(t)}{\partial \lambda} \right|_{\lambda=0} = - \sum_{\tau} \langle O_\pi(t) \mathcal{O}(\tau) O_\pi^\dagger(0) \rangle, \quad (9)$$

up to the sign convention chosen in the action perturbation. In the code-sign convention of Eq. (7), the response contraction is compared to

$$\sum_{\tau} C_3(t, \tau). \quad (10)$$

This explains why a linearized background-response calculation has analysis features similar to a summed three-point function: it is the derivative  $\partial C_\lambda / \partial \lambda|_0$ , not a finite- $\lambda$  energy fit.

## 3 Pion EMFF Kinematics

The pion EMFF matrix element is defined by

$$\langle \pi(p_f) | J_\mu(0) | \pi(p_i) \rangle = (p_i + p_f)_\mu F_\pi(Q^2), \quad q = p_f - p_i. \quad (11)$$

The lattice implementation uses

$$\mathbf{p}_i = \mathbf{p}_f - \mathbf{q}_{\text{ext}}. \quad (12)$$

The explicit three-point correlator is

$$C_3^\Gamma(\mathbf{q}, \tau; \mathbf{p}_f, t_{\text{sep}}) = \sum_{\mathbf{x}} e^{-i\mathbf{q} \cdot (\mathbf{x} - \mathbf{x}_0)} \text{tr} [S_{\text{seq,anti}}(\mathbf{x}, \tau; \mathbf{p}_f, t_{\text{sep}}) \Gamma S_q(\mathbf{x}, \tau; \mathbf{x}_0, 0) \Gamma_{\text{src}}], \quad (13)$$

where the fixed-sink sequential propagator has already absorbed the sink momentum  $\mathbf{p}_f$ , the sink time  $t_{\text{sep}}$ , and the sink interpolator  $\Gamma_{\text{sink}}$ . The standard EMFF current is the temporal vector current,

$$\Gamma = \gamma_T, \quad (14)$$

labeled T in the code.

## 4 Current-Inserted Source

For a local current with momentum transfer  $\mathbf{q}$ ,

$$\mathcal{O}_{\Gamma, \mathbf{q}}(x) = \Gamma \Phi_{\mathbf{q}}(x - \mathbf{x}_0), \quad (15)$$

where  $\Phi_{\mathbf{q}}$  is produced by `MomentumPhase` using the same phase convention as the ordinary EMFF code.

Given an ordinary source-to-current propagator

$$S(x, x_0), \quad (16)$$

the all-time current-inserted source is

$$\eta_{\Gamma, \mathbf{q}}(x) = \Gamma \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0). \quad (17)$$

The response propagator is then obtained by a standard inversion,

$$S_{\text{resp}}(y, x_0; \Gamma, \mathbf{q}) = \sum_x S(y, x) \Gamma \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0). \quad (18)$$

In code, Eq. (17) is built by

```
build_local_current_inserted_source(...)
```

and Eq. (18) by

```
invert_local_current_response_propagator(...).
```

## 5 Response Correlator

The response propagator can be contracted like the forward quark line in a pion two-point function. With an ordinary antiquark line  $S_{\text{anti}}$ ,

$$C_{\text{resp}}(t; \mathbf{p}_f, \mathbf{q}) = \sum_{\mathbf{y}} e^{-i\mathbf{p}_f \cdot (\mathbf{y} - \mathbf{x}_0)} \text{tr} [S_{\text{anti}}(\mathbf{y}, t; \mathbf{x}_0, 0) \Gamma_{\text{sink}} S_{\text{resp}}(\mathbf{y}, t; \mathbf{x}_0, 0; \Gamma, \mathbf{q}) \Gamma_{\text{src}}]. \quad (19)$$

For the code-sign convention,

$$C_{\text{resp}}(t_{\text{sep}}; \mathbf{p}_f, \mathbf{q}) = \sum_{\tau} C_3^{\Gamma}(\mathbf{q}, \tau; \mathbf{p}_f, t_{\text{sep}}), \quad (20)$$

provided the same current gamma, source/sink interpolators, momentum phases, and smearing conventions are used.

The first validated case is

$$\mathbf{p}_f = \mathbf{0}, \quad \mathbf{q} = \mathbf{0}, \quad \mathbf{p}_i = \mathbf{0}, \quad \Gamma = \gamma_T, \quad \Gamma_{\text{src}} = \Gamma_{\text{sink}} = \gamma_5. \quad (21)$$

On the S8T32 test gauge, the diagnostic found

$$\sum_{\tau} C_3(\tau) = 327.27206385480486 + 0.1342484007638715 i, \quad (22)$$

$$C_{\text{resp}} = 327.27206385480486 + 0.13424840076386432 i, \quad (23)$$

with relative difference

$$2.2 \times 10^{-17}. \quad (24)$$

## 6 Energy-Shift Interpretation

Although Eq. (20) looks like a summed three-point function, it is still directly related to the background-field energy shift. Suppose

$$C_{\lambda}(t) = Z_{\lambda} e^{-E_{\lambda} t}. \quad (25)$$

Expanding around  $\lambda = 0$ ,

$$Z_{\lambda} = Z_0 + \lambda Z'_0 + O(\lambda^2), \quad (26)$$

$$E_{\lambda} = E_0 + \lambda E'_0 + O(\lambda^2), \quad (27)$$

gives

$$C_{\lambda}(t) = C_0(t) \left[ 1 + \lambda \left( \frac{Z'_0}{Z_0} - t E'_0 \right) \right] + O(\lambda^2). \quad (28)$$

Therefore

$$\frac{1}{C_0(t)} \left. \frac{\partial C_{\lambda}(t)}{\partial \lambda} \right|_{\lambda=0} = \frac{Z'_0}{Z_0} - t E'_0. \quad (29)$$

Thus, in the linear-response method, the energy-shift derivative is extracted from the slope of a response ratio, not from separate finite- $\lambda$  energy fits. This is the same Feynman-Hellmann information, but represented at  $\lambda = 0$ .

## 7 Summed-Ratio Object

Define

$$R_{\text{sum}}(t_{\text{sep}}) = \frac{\sum_{\tau} C_3(t_{\text{sep}}, \tau)}{C_2(t_{\text{sep}})} = \frac{C_{\text{resp}}(t_{\text{sep}})}{C_2(t_{\text{sep}})}. \quad (30)$$

At large  $t_{\text{sep}}$ ,

$$R_{\text{sum}}(t_{\text{sep}}) = A - t_{\text{sep}} E'_0 + \text{excited-state corrections}, \quad (31)$$

where  $A$  absorbs overlap-factor response and contact-term effects.

For the finite- $\lambda$  sign convention  $D_{\lambda} = D + \lambda \mathcal{O}$ , the derivative of the propagator has the sign in Eq. (6). The diagnostic code stores both

$$\text{response\_sign} = +1$$

for direct comparison to the explicit summed three-point function, and

$$\text{finite\_difference\_derivative\_sign} = -1$$

for the corresponding finite-field derivative convention.

## 8 Restricted Insertion-Time Windows

The full background-field energy-shift picture corresponds to an all-time insertion. However, in the linearized response implementation one may restrict the insertion time,

$$S_{\text{resp}}^{W_{\tau}} = \sum_{\tau \in W_{\tau}} D^{-1} [\delta_{t,\tau} \Gamma \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0)]. \quad (32)$$

This produces

$$C_{\text{resp}}^{W_{\tau}}(t_{\text{sep}}) = \sum_{\tau \in W_{\tau}} C_3(t_{\text{sep}}, \tau). \quad (33)$$

Useful choices include

$$W_{\tau} = \text{all time slices}, \quad (34)$$

$$W_{\tau} = \{0, \dots, t_{\text{sep}}\}, \quad (35)$$

$$W_{\tau} = \{1, \dots, t_{\text{sep}} - 1\}, \quad (36)$$

$$W_{\tau} = \{\tau_{\text{min}}, \dots, t_{\text{sep}} - \tau_{\text{min}}\}. \quad (37)$$

The restricted windows are not the pure finite-field energy-shift method. They are analysis tools for the linearized method, useful for:

- removing source/sink contact terms,
- comparing with explicit three-point functions in the same time window,
- studying excited-state contamination,
- checking the stability of summed-ratio slopes.

## 9 Current Implementation

### 9.1 Source helper

The helper

```
build_local_current_inserted_source(prop_forward, phase_q, current_gamma, tau)
```

constructs

$$\eta(x) = \Gamma_{\text{cur}} \Phi_{\mathbf{q}}(x - \mathbf{x}_0) S(x, x_0). \quad (38)$$

If `tau` is provided, the source is masked to that global insertion time. If `tau` is omitted, all time slices are included.

### 9.2 Response inversion

The helper

```
invert_local_current_response_propagator(...)
```

performs the ordinary  $D^{-1}$  inversion of the current-inserted source. If a list of times `tau_list` is given, it inverts one time-sliced source per insertion time and sums the resulting propagators.

### 9.3 Response contraction

The helper

```
contract_response_pion_2pt(...)
```

contracts the response propagator with the ordinary antiquark line using the same pion two-point contraction infrastructure as the rest of the pion code.

### 9.4 Diagnostic application

The Perlmutter diagnostic application is

```
application/EMFF_pion_background_response/perlmutter/.
```

The default run uses the S8T32 test gauge and compares

$$C_{\text{resp}}(t_{\text{sep}}) \quad \text{against} \quad \sum_{\tau} C_3(t_{\text{sep}}, \tau) \quad (39)$$

for  $t_{\text{sep}} = 2$ ,  $\Gamma = \gamma_T$ , and zero external momentum.

## 10 Recommended Upgrade Path

1. Validate nonzero momentum transfer:

$$\mathbf{p}_f = (0, 0, 1), \quad \mathbf{q} = (0, 0, 1), \quad \mathbf{p}_i = (0, 0, 0).$$

Then validate Breit-frame kinematics:

$$\mathbf{p}_f = (0, 0, 1), \quad \mathbf{q} = (0, 0, 2), \quad \mathbf{p}_i = (0, 0, -1).$$

2. Add multi- $t_{\text{sep}}$  output and save

$$C_2(t_{\text{sep}}), \quad C_{\text{resp}}(t_{\text{sep}}), \quad R_{\text{sum}}(t_{\text{sep}}).$$

3. Add restricted time windows:

$$W_\tau = \{\tau_{\min}, \dots, t_{\text{sep}} - \tau_{\min}\}.$$

4. Cache per- $\tau$  response propagators:

$$\phi_\tau = D^{-1}[\delta_{t,\tau}\Gamma\Phi_{\mathbf{q}}S],$$

then build arbitrary windows by summing  $\phi_\tau$ .

5. Extend from the temporal vector current to a bilinear response suite: scalar, vector, axial, pseudoscalar, and tensor currents.
6. Prototype straight Wilson-line PDF response:

$$\mathcal{O}_z(x) = \bar{q}(x+z)\Gamma W_z(x+z, x)q(x),$$

again without modifying the Dirac operator.

## 11 Summary

The present pion EMFF background-response code is a linearized Feynman-Hellmann diagnostic:

$$S_{\text{resp}} = D^{-1}\Gamma\Phi_{\mathbf{q}}S. \tag{40}$$

It does not compute finite- $\lambda$  energies. Instead, it computes  $\partial C_\lambda/\partial\lambda|_{\lambda=0}$ , which equals a summed three-point function up to the convention-dependent sign. The energy-shift derivative is then encoded in the slope of the response ratio  $C_{\text{resp}}/C_2$  as a function of  $t_{\text{sep}}$ .